Blackmon Lab

Evolutionary Biology

Quantifying Proteome Space Variation in Domesticated and Wild Carnivores Using Protein Language Model

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Background

Domestication has produced major changes in animal morphology, behavior, and genetics, yet its effects on proteome-wide variation remain poorly understood. One key question is whether domesticated species explore a different or expanded region of "protein space," a high-dimensional representation of all possible protein sequences and their biophysical properties.

Two opposing evolutionary forces shape this process. Domestication introduces population bottlenecks that reduce genetic diversity, potentially contracting protein space. At the same time, relaxed selection in human-managed environments may allow slightly deleterious or structurally diverse proteins to persist, potentially expanding protein space.

In this study, we compare domesticated species (dog and chicken) with their wild relatives (wolf and red junglefowl) to test whether domestication alters the structure and volume of proteome space. Protein sequences were analyzed using ESM-2 embeddings, which represent each proteome as a distribution of points in high-dimensional space. Convex hull volume was used as a preliminary measure of proteome spread.

This approach provides a novel framework for evaluating how evolutionary processes shape proteome diversity at the molecular level.

Abstract

Domestication alters evolutionary pressures, but its effects on proteome-wide diversity remain unclear. We compared domesticated species (dog and chicken) with their wild relatives (wolf and red junglefowl) using ESM-2 protein embeddings to represent proteomes in high-dimensional space. Convex hull volume was used as a preliminary measure of proteome spread. Results suggest that domesticated species exhibit differences in protein space structure, supporting the idea that domestication influences molecular diversity.

Study Design & AI

I compared domesticated species (dog and chicken) with their wild relatives (wolf and red junglefowl) to assess the effects of domestication on proteome space. Protein sequences were obtained from public databases and filtered for quality and redundancy. Each proteome was treated as a representation of its species' protein space.

Protein sequences were converted into embeddings using ESM-2. Principal Component Analysis (PCA) was applied for dimensionality reduction. Proteome spread was quantified using convex hull volume, and differences between domesticated and wild species were used to evaluate changes in protein space. AI was also used to help with python scripts.

Objectives and Hypothesis

•Objectives

•Compare proteome variation between domesticated and wild carnivores.

•Quantify proteome "space" using protein embedding methods.

•Build a reproducible workflow for comparing species at the proteome level.

•Hypothesis

•Domesticated carnivores will show altered proteome diversity relative to wild relatives because artificial selection may shift the structure of protein space.

Materials and Methods

•Selected carnivore species dog vs wolf and chicken vs junglefowl
•Downloaded proteomes from NCBI datasets
•Generated protein embeddings using ESM-2 model
•Represented proteomes as point clouds in high-dimensional space
•Applied dimensionality reduction (e.g., PCA)
•Quantified spread/volume of embedding distributions

Results

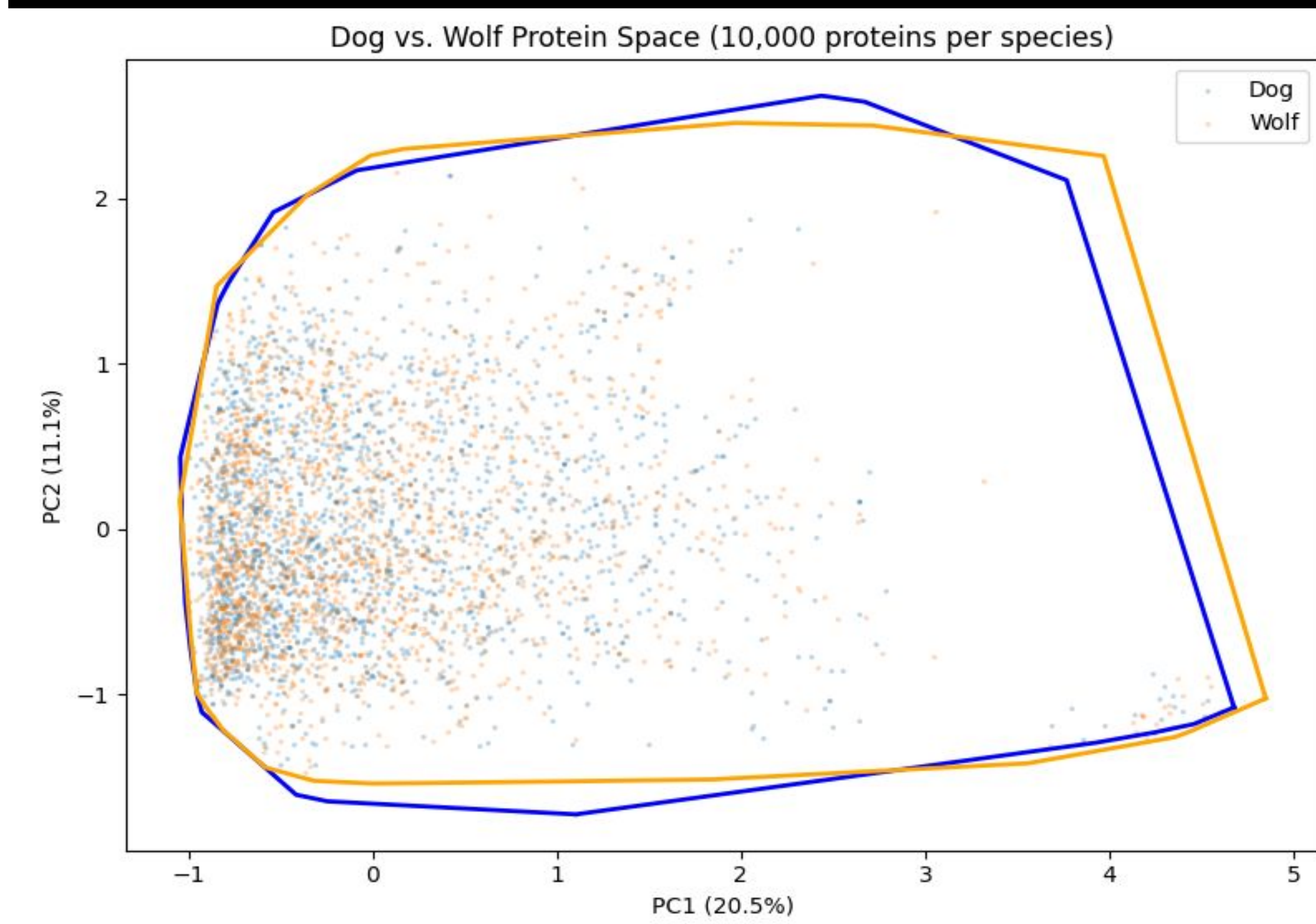

Figure 1. PCA plot of dog and wolf protein embeddings (10,000 proteins/species). Points represent proteins; convex hulls show overall proteome spread. Substantial overlap is observed. *(generated in Python)*

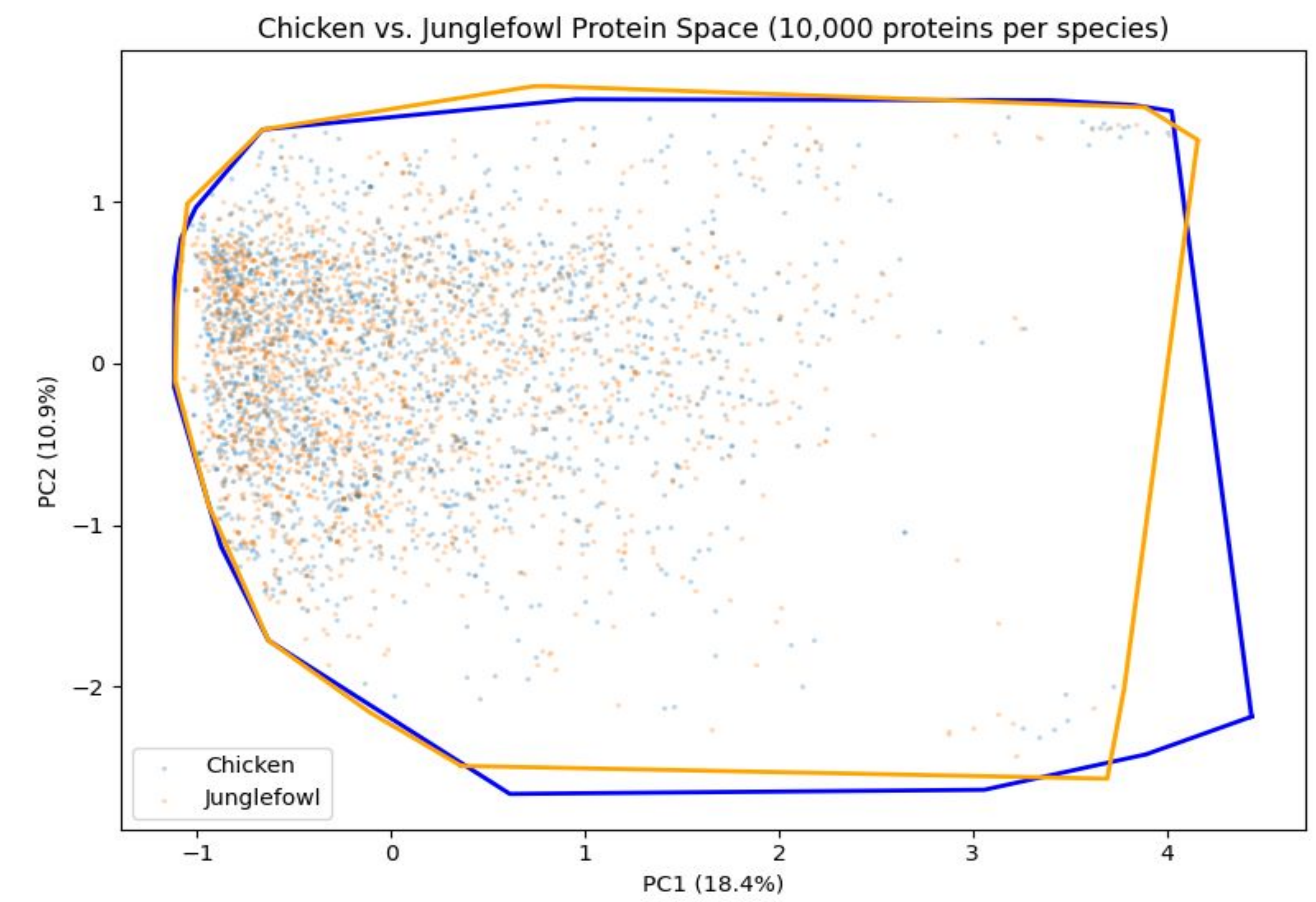

Figure 2. PCA plot of chicken and red junglefowl protein embeddings (10,000 proteins/species). Junglefowl shows a larger protein space than chicken. *(generated in Python)*

To ensure robustness, multiple trials were conducted using different random samples of proteins. In the chicken–red junglefowl comparison, results were highly consistent across all replicates, with wild junglefowl exhibiting a larger protein space in every trial. In contrast, the dog–wolf comparison showed more variability across replicates, with no consistent pattern in which species had a larger protein space.

Discussion

The chicken–red junglefowl comparison provides strong evidence that domestication can reduce protein space. Across six independent replicates, the wild junglefowl consistently exhibited a larger convex hull volume (40.01 ± 2.60) than domesticated chickens (33.57 ± 2.98), representing an approximately 19% increase in protein space. The consistency of this result, with no overlap across replicates, indicates a robust difference in proteome structure between wild and domesticated lineages.

This pattern supports the hypothesis that domestication can constrain protein diversity, likely due to population bottlenecks and strong artificial selection. In chickens, selective breeding for traits such as egg production and growth may have reduced overall genetic and functional variation, leading to a smaller protein space compared to their wild ancestor.

In contrast, results from the dog–wolf comparison were less consistent, suggesting that domestication does not produce uniform effects across species. Differences in evolutionary history, population structure, and gene flow may explain this variation. For example, dogs may experience greater diversification due to breed formation and occasional introgression from wild populations, while chickens may be subject to more intense directional selection.

Despite the strength of the chicken–junglefowl result, some limitations remain. Convex hull volume can be sensitive to outliers and sampling effects, and further analyses using additional metrics and larger sample sizes are needed to confirm these patterns.

Overall, these findings demonstrate that domestication can significantly alter protein space, but the direction and magnitude of this effect depend on species-specific evolutionary dynamics.

Conclusions & Acknowledgements

This study shows that protein language models can be used to measure and compare proteome diversity. Results suggest that domestication can affect protein space, with a clear reduction observed in chickens, while patterns in dogs were less consistent. Overall, these findings highlight the complexity of domestication and the value of computational methods in studying molecular variation.

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